



Build a Fair Probability Model Teaching Bundle

Grade 7 • North Carolina • Mathematics

NC.7.SP.7 • Use equally likely outcomes to model chance.

A focused classroom collection: task cards, five warm-ups, classwork, an exit ticket, four seasonal practice sets, substitute lessons, and planning pages. Each section has worked answers.

PRINT • TEACH • REVISIT



Your teaching route

Focus

Use equally likely outcomes to model chance. This bundle practices a focused part of NC.7.SP.7; it is not a complete grade curriculum or a simulated EOG.

Before teaching

Learners should explain a simpler example using a model. Read the two worked examples, then use the first task card to check readiness.

Suggested sequence

Day 1: model for 10 minutes, then guided task cards for 15 minutes. Day 2: a 5-minute warm-up and 20 minutes of practice. Use the exit ticket to choose a small-group reteaching focus. Seasonal sets revisit the same skill using different quantities.

Materials

Pencils, blank paper, and graph paper or counters as useful. No paid app or internet is needed for student work. A calculator may check finished calculations; these are instructional choices, not official testing accommodation rules.



Make the reasoning visible

List the equally likely outcomes. Probability is favorable outcomes divided by all outcomes. Check that probabilities total one.

Example 1: A bag holds 3 red, 4 blue, and 3 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

There are 10 equally likely counters. Red: $3/10$; not red: $7/10$. These disjoint events cover all 10 counters.

Example 2: A bag holds 3 red, 7 blue, and 5 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

There are 15 equally likely counters. Red: $3/15$; not red: $12/15$. These disjoint events cover all 15 counters.

Ask: What does each number represent? How could a model or inverse operation check the result?

Task cards 1–4

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 01

A bag holds 8 red, 3 blue, and 7 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

Explain your method.

CARD 02

A bag holds 3 red, 3 blue, and 7 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

Explain your method.

CARD 03

A bag holds 2 red, 8 blue, and 7 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

Explain your method.

CARD 04

A bag holds 3 red, 8 blue, and 3 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

Explain your method.

Task cards 5–8

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 05

A bag holds 8 red, 3 blue, and 3 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

Explain your method.

CARD 06

A bag holds 2 red, 8 blue, and 6 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

Explain your method.

CARD 07

A bag holds 5 red, 2 blue, and 2 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

Explain your method.

CARD 08

A bag holds 5 red, 8 blue, and 7 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

Explain your method.

Task cards 9–12

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 09

A bag holds 6 red, 8 blue, and 3 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

Explain your method.

CARD 10

A bag holds 2 red, 2 blue, and 7 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

Explain your method.

CARD 11

A bag holds 8 red, 5 blue, and 4 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

Explain your method.

CARD 12

A bag holds 2 red, 2 blue, and 7 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

Explain your method.

Task cards 13–16

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 13

A bag holds 4 red, 3 blue, and 5 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

Explain your method.

CARD 14

A bag holds 7 red, 5 blue, and 3 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

Explain your method.

CARD 15

A bag holds 4 red, 2 blue, and 4 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

Explain your method.

CARD 16

A bag holds 5 red, 2 blue, and 7 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

Explain your method.

Task-card response record

Use this page for final answers; attach drawings or full worked solutions on separate paper.

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____
9. _____
10. _____
11. _____
12. _____
13. _____
14. _____
15. _____
16. _____

Warm-up • Day 1

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

1. A bag holds 2 red, 5 blue, and 7 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

2. A bag holds 2 red, 8 blue, and 3 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.



Warm-up • Day 2

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

3. A bag holds 5 red, 7 blue, and 2 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

4. A bag holds 3 red, 3 blue, and 3 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.



Warm-up • Day 3

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

5. A bag holds 7 red, 7 blue, and 4 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

6. A bag holds 8 red, 5 blue, and 2 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.



Warm-up • Day 4

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

7. A bag holds 3 red, 2 blue, and 7 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

8. A bag holds 7 red, 7 blue, and 3 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.



Warm-up • Day 5

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

9. A bag holds 7 red, 5 blue, and 6 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

10. A bag holds 5 red, 7 blue, and 6 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.



Classwork • Represent and explain

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. A bag holds 7 red, 3 blue, and 3 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

2. A bag holds 4 red, 7 blue, and 6 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

3. A bag holds 3 red, 8 blue, and 2 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.



Classwork • Represent and explain

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

4. A bag holds 4 red, 5 blue, and 4 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

5. A bag holds 6 red, 6 blue, and 4 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

6. A bag holds 4 red, 2 blue, and 2 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.



Exit ticket • Try independently

Complete the four questions independently. Show enough reasoning for your teacher to see your method. This is a classroom check, not a standardized score.

1. A bag holds 7 red, 3 blue, and 2 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

2. A bag holds 5 red, 6 blue, and 3 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

3. A bag holds 4 red, 6 blue, and 4 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

4. A bag holds 2 red, 2 blue, and 4 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.



Task-card worked solutions

1. $\frac{4}{9}$; $\frac{5}{9}$

There are 18 equally likely counters. Red: $\frac{8}{18}$; not red: $\frac{10}{18}$. These disjoint events cover all 18 counters.

2. $\frac{3}{13}$; $\frac{10}{13}$

There are 13 equally likely counters. Red: $\frac{3}{13}$; not red: $\frac{10}{13}$. These disjoint events cover all 13 counters.

3. $\frac{2}{17}$; $\frac{15}{17}$

There are 17 equally likely counters. Red: $\frac{2}{17}$; not red: $\frac{15}{17}$. These disjoint events cover all 17 counters.

4. $\frac{3}{14}$; $\frac{11}{14}$

There are 14 equally likely counters. Red: $\frac{3}{14}$; not red: $\frac{11}{14}$. These disjoint events cover all 14 counters.

5. $\frac{4}{7}$; $\frac{3}{7}$

There are 14 equally likely counters. Red: $\frac{8}{14}$; not red: $\frac{6}{14}$. These disjoint events cover all 14 counters.

6. $\frac{1}{8}$; $\frac{7}{8}$

There are 16 equally likely counters. Red: $\frac{2}{16}$; not red: $\frac{14}{16}$. These disjoint events cover all 16 counters.

Task-card worked solutions

7. $\frac{5}{9}$; $\frac{4}{9}$

There are 9 equally likely counters. Red: $\frac{5}{9}$; not red: $\frac{4}{9}$. These disjoint events cover all 9 counters.

8. $\frac{1}{4}$; $\frac{3}{4}$

There are 20 equally likely counters. Red: $\frac{5}{20}$; not red: $\frac{15}{20}$. These disjoint events cover all 20 counters.

9. $\frac{6}{17}$; $\frac{11}{17}$

There are 17 equally likely counters. Red: $\frac{6}{17}$; not red: $\frac{11}{17}$. These disjoint events cover all 17 counters.

10. $\frac{2}{11}$; $\frac{9}{11}$

There are 11 equally likely counters. Red: $\frac{2}{11}$; not red: $\frac{9}{11}$. These disjoint events cover all 11 counters.

11. $\frac{8}{17}$; $\frac{9}{17}$

There are 17 equally likely counters. Red: $\frac{8}{17}$; not red: $\frac{9}{17}$. These disjoint events cover all 17 counters.

12. $\frac{2}{11}$; $\frac{9}{11}$

There are 11 equally likely counters. Red: $\frac{2}{11}$; not red: $\frac{9}{11}$. These disjoint events cover all 11 counters.

Task-card worked solutions

13. $\frac{1}{3}$; $\frac{2}{3}$

There are 12 equally likely counters. Red: $\frac{4}{12}$; not red: $\frac{8}{12}$. These disjoint events cover all 12 counters.

14. $\frac{7}{15}$; $\frac{8}{15}$

There are 15 equally likely counters. Red: $\frac{7}{15}$; not red: $\frac{8}{15}$. These disjoint events cover all 15 counters.

15. $\frac{2}{5}$; $\frac{3}{5}$

There are 10 equally likely counters. Red: $\frac{4}{10}$; not red: $\frac{6}{10}$. These disjoint events cover all 10 counters.

16. $\frac{5}{14}$; $\frac{9}{14}$

There are 14 equally likely counters. Red: $\frac{5}{14}$; not red: $\frac{9}{14}$. These disjoint events cover all 14 counters.

Warm-up worked solutions

1. $\frac{1}{7}$; $\frac{6}{7}$

There are 14 equally likely counters. Red: $\frac{2}{14}$; not red: $\frac{12}{14}$. These disjoint events cover all 14 counters.

2. $\frac{2}{13}$; $\frac{11}{13}$

There are 13 equally likely counters. Red: $\frac{2}{13}$; not red: $\frac{11}{13}$. These disjoint events cover all 13 counters.

3. $\frac{5}{14}$; $\frac{9}{14}$

There are 14 equally likely counters. Red: $\frac{5}{14}$; not red: $\frac{9}{14}$. These disjoint events cover all 14 counters.

4. $\frac{1}{3}$; $\frac{2}{3}$

There are 9 equally likely counters. Red: $\frac{3}{9}$; not red: $\frac{6}{9}$. These disjoint events cover all 9 counters.

5. $\frac{7}{18}$; $\frac{11}{18}$

There are 18 equally likely counters. Red: $\frac{7}{18}$; not red: $\frac{11}{18}$. These disjoint events cover all 18 counters.

6. $\frac{8}{15}$; $\frac{7}{15}$

There are 15 equally likely counters. Red: $\frac{8}{15}$; not red: $\frac{7}{15}$. These disjoint events cover all 15 counters.

Warm-up worked solutions

7. $\frac{1}{4}$; $\frac{3}{4}$

There are 12 equally likely counters. Red: $\frac{3}{12}$; not red: $\frac{9}{12}$. These disjoint events cover all 12 counters.

8. $\frac{7}{17}$; $\frac{10}{17}$

There are 17 equally likely counters. Red: $\frac{7}{17}$; not red: $\frac{10}{17}$. These disjoint events cover all 17 counters.

9. $\frac{7}{18}$; $\frac{11}{18}$

There are 18 equally likely counters. Red: $\frac{7}{18}$; not red: $\frac{11}{18}$. These disjoint events cover all 18 counters.

10. $\frac{5}{18}$; $\frac{13}{18}$

There are 18 equally likely counters. Red: $\frac{5}{18}$; not red: $\frac{13}{18}$. These disjoint events cover all 18 counters.

Classwork worked solutions

1. $\frac{7}{13}$; $\frac{6}{13}$

There are 13 equally likely counters. Red: $\frac{7}{13}$; not red: $\frac{6}{13}$. These disjoint events cover all 13 counters.

2. $\frac{4}{17}$; $\frac{13}{17}$

There are 17 equally likely counters. Red: $\frac{4}{17}$; not red: $\frac{13}{17}$. These disjoint events cover all 17 counters.

3. $\frac{3}{13}$; $\frac{10}{13}$

There are 13 equally likely counters. Red: $\frac{3}{13}$; not red: $\frac{10}{13}$. These disjoint events cover all 13 counters.

4. $\frac{4}{13}$; $\frac{9}{13}$

There are 13 equally likely counters. Red: $\frac{4}{13}$; not red: $\frac{9}{13}$. These disjoint events cover all 13 counters.

5. $\frac{3}{8}$; $\frac{5}{8}$

There are 16 equally likely counters. Red: $\frac{6}{16}$; not red: $\frac{10}{16}$. These disjoint events cover all 16 counters.

6. $\frac{1}{2}$; $\frac{1}{2}$

There are 8 equally likely counters. Red: $\frac{4}{8}$; not red: $\frac{4}{8}$. These disjoint events cover all 8 counters.

Exit-ticket worked solutions

1. $\frac{7}{12}$; $\frac{5}{12}$

There are 12 equally likely counters. Red: $\frac{7}{12}$; not red: $\frac{5}{12}$. These disjoint events cover all 12 counters.

2. $\frac{5}{14}$; $\frac{9}{14}$

There are 14 equally likely counters. Red: $\frac{5}{14}$; not red: $\frac{9}{14}$. These disjoint events cover all 14 counters.

3. $\frac{2}{7}$; $\frac{5}{7}$

There are 14 equally likely counters. Red: $\frac{4}{14}$; not red: $\frac{10}{14}$. These disjoint events cover all 14 counters.

4. $\frac{1}{4}$; $\frac{3}{4}$

There are 8 equally likely counters. Red: $\frac{2}{8}$; not red: $\frac{6}{8}$. These disjoint events cover all 8 counters.

Fall • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. A bag holds 3 red, 6 blue, and 5 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

2. A bag holds 2 red, 2 blue, and 2 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

3. A bag holds 8 red, 7 blue, and 3 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

4. A bag holds 7 red, 7 blue, and 7 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.



Fall • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. A bag holds 5 red, 4 blue, and 6 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

6. A bag holds 6 red, 6 blue, and 4 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

7. A bag holds 3 red, 8 blue, and 6 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

8. A bag holds 9 red, 3 blue, and 6 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.



Fall • Worked solutions

1. $\frac{3}{14}$; $\frac{11}{14}$

There are 14 equally likely counters. Red: $\frac{3}{14}$; not red: $\frac{11}{14}$. These disjoint events cover all 14 counters.

2. $\frac{1}{3}$; $\frac{2}{3}$

There are 6 equally likely counters. Red: $\frac{2}{6}$; not red: $\frac{4}{6}$. These disjoint events cover all 6 counters.

3. $\frac{4}{9}$; $\frac{5}{9}$

There are 18 equally likely counters. Red: $\frac{8}{18}$; not red: $\frac{10}{18}$. These disjoint events cover all 18 counters.

4. $\frac{1}{3}$; $\frac{2}{3}$

There are 21 equally likely counters. Red: $\frac{7}{21}$; not red: $\frac{14}{21}$. These disjoint events cover all 21 counters.

Fall • Worked solutions

5. $\frac{1}{3}$; $\frac{2}{3}$

There are 15 equally likely counters. Red: $\frac{5}{15}$; not red: $\frac{10}{15}$. These disjoint events cover all 15 counters.

6. $\frac{3}{8}$; $\frac{5}{8}$

There are 16 equally likely counters. Red: $\frac{6}{16}$; not red: $\frac{10}{16}$. These disjoint events cover all 16 counters.

7. $\frac{3}{17}$; $\frac{14}{17}$

There are 17 equally likely counters. Red: $\frac{3}{17}$; not red: $\frac{14}{17}$. These disjoint events cover all 17 counters.

8. $\frac{1}{2}$; $\frac{1}{2}$

There are 18 equally likely counters. Red: $\frac{9}{18}$; not red: $\frac{9}{18}$. These disjoint events cover all 18 counters.

Winter • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. A bag holds 2 red, 7 blue, and 7 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

2. A bag holds 3 red, 7 blue, and 6 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

3. A bag holds 4 red, 8 blue, and 2 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

4. A bag holds 7 red, 7 blue, and 3 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.



Winter • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. A bag holds 3 red, 5 blue, and 4 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

6. A bag holds 6 red, 7 blue, and 2 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

7. A bag holds 7 red, 5 blue, and 7 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

8. A bag holds 9 red, 3 blue, and 4 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.



Winter • Worked solutions

1. $\frac{1}{8}$; $\frac{7}{8}$

There are 16 equally likely counters. Red: $\frac{2}{16}$; not red: $\frac{14}{16}$. These disjoint events cover all 16 counters.

2. $\frac{3}{16}$; $\frac{13}{16}$

There are 16 equally likely counters. Red: $\frac{3}{16}$; not red: $\frac{13}{16}$. These disjoint events cover all 16 counters.

3. $\frac{2}{7}$; $\frac{5}{7}$

There are 14 equally likely counters. Red: $\frac{4}{14}$; not red: $\frac{10}{14}$. These disjoint events cover all 14 counters.

4. $\frac{7}{17}$; $\frac{10}{17}$

There are 17 equally likely counters. Red: $\frac{7}{17}$; not red: $\frac{10}{17}$. These disjoint events cover all 17 counters.

Winter • Worked solutions

5. $\frac{1}{4}$; $\frac{3}{4}$

There are 12 equally likely counters. Red: $\frac{3}{12}$; not red: $\frac{9}{12}$. These disjoint events cover all 12 counters.

6. $\frac{2}{5}$; $\frac{3}{5}$

There are 15 equally likely counters. Red: $\frac{6}{15}$; not red: $\frac{9}{15}$. These disjoint events cover all 15 counters.

7. $\frac{7}{19}$; $\frac{12}{19}$

There are 19 equally likely counters. Red: $\frac{7}{19}$; not red: $\frac{12}{19}$. These disjoint events cover all 19 counters.

8. $\frac{9}{16}$; $\frac{7}{16}$

There are 16 equally likely counters. Red: $\frac{9}{16}$; not red: $\frac{7}{16}$. These disjoint events cover all 16 counters.

Spring break • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. A bag holds 8 red, 7 blue, and 6 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

2. A bag holds 2 red, 5 blue, and 5 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

3. A bag holds 7 red, 3 blue, and 7 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

4. A bag holds 2 red, 2 blue, and 4 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.



Spring break • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. A bag holds 6 red, 5 blue, and 4 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

6. A bag holds 7 red, 5 blue, and 5 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

7. A bag holds 8 red, 8 blue, and 4 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

8. A bag holds 7 red, 7 blue, and 3 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.



Spring break • Worked solutions

1. $\frac{8}{21}$; $\frac{13}{21}$

There are 21 equally likely counters. Red: $\frac{8}{21}$; not red: $\frac{13}{21}$. These disjoint events cover all 21 counters.

2. $\frac{1}{6}$; $\frac{5}{6}$

There are 12 equally likely counters. Red: $\frac{2}{12}$; not red: $\frac{10}{12}$. These disjoint events cover all 12 counters.

3. $\frac{7}{17}$; $\frac{10}{17}$

There are 17 equally likely counters. Red: $\frac{7}{17}$; not red: $\frac{10}{17}$. These disjoint events cover all 17 counters.

4. $\frac{1}{4}$; $\frac{3}{4}$

There are 8 equally likely counters. Red: $\frac{2}{8}$; not red: $\frac{6}{8}$. These disjoint events cover all 8 counters.

Spring break • Worked solutions

5. $\frac{2}{5}$; $\frac{3}{5}$

There are 15 equally likely counters. Red: $\frac{6}{15}$; not red: $\frac{9}{15}$. These disjoint events cover all 15 counters.

6. $\frac{7}{17}$; $\frac{10}{17}$

There are 17 equally likely counters. Red: $\frac{7}{17}$; not red: $\frac{10}{17}$. These disjoint events cover all 17 counters.

7. $\frac{2}{5}$; $\frac{3}{5}$

There are 20 equally likely counters. Red: $\frac{8}{20}$; not red: $\frac{12}{20}$. These disjoint events cover all 20 counters.

8. $\frac{7}{17}$; $\frac{10}{17}$

There are 17 equally likely counters. Red: $\frac{7}{17}$; not red: $\frac{10}{17}$. These disjoint events cover all 17 counters.

Summer • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. A bag holds 7 red, 7 blue, and 2 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

2. A bag holds 4 red, 4 blue, and 3 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

3. A bag holds 3 red, 7 blue, and 2 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

4. A bag holds 8 red, 5 blue, and 7 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.



Summer • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. A bag holds 8 red, 4 blue, and 5 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

6. A bag holds 9 red, 4 blue, and 3 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

7. A bag holds 6 red, 7 blue, and 7 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

8. A bag holds 4 red, 7 blue, and 7 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.



Summer • Worked solutions

1. $\frac{7}{16}$; $\frac{9}{16}$

There are 16 equally likely counters. Red: $\frac{7}{16}$; not red: $\frac{9}{16}$. These disjoint events cover all 16 counters.

2. $\frac{4}{11}$; $\frac{7}{11}$

There are 11 equally likely counters. Red: $\frac{4}{11}$; not red: $\frac{7}{11}$. These disjoint events cover all 11 counters.

3. $\frac{1}{4}$; $\frac{3}{4}$

There are 12 equally likely counters. Red: $\frac{3}{12}$; not red: $\frac{9}{12}$. These disjoint events cover all 12 counters.

4. $\frac{2}{5}$; $\frac{3}{5}$

There are 20 equally likely counters. Red: $\frac{8}{20}$; not red: $\frac{12}{20}$. These disjoint events cover all 20 counters.

Summer • Worked solutions

5. $\frac{8}{17}$; $\frac{9}{17}$

There are 17 equally likely counters. Red: $\frac{8}{17}$; not red: $\frac{9}{17}$. These disjoint events cover all 17 counters.

6. $\frac{9}{16}$; $\frac{7}{16}$

There are 16 equally likely counters. Red: $\frac{9}{16}$; not red: $\frac{7}{16}$. These disjoint events cover all 16 counters.

7. $\frac{3}{10}$; $\frac{7}{10}$

There are 20 equally likely counters. Red: $\frac{6}{20}$; not red: $\frac{14}{20}$. These disjoint events cover all 20 counters.

8. $\frac{2}{9}$; $\frac{7}{9}$

There are 18 equally likely counters. Red: $\frac{4}{18}$; not red: $\frac{14}{18}$. These disjoint events cover all 18 counters.

Substitute lesson

Before students arrive

Locate the class schedule, seating plan, emergency procedures, and approved learner supports from the school. This academic packet does not replace those school-specific instructions.

0-5 minutes: settle

Read the goal aloud: Use equally likely outcomes to model chance. Let students explain a familiar strategy to a partner.

5-15 minutes: model

Use the worked-example page. List the equally likely outcomes. Probability is favorable outcomes divided by all outcomes. Check that probabilities total one.

15-30 minutes: practice

Distribute the next practice page. Students solve independently first, then compare one explanation in pairs. If copying is unavailable, read one question at a time and allow drawings or oral responses.

30-40 minutes: discuss

Use the following key. Ask students to locate the first step where methods diverged; invite a correction supported by a model.

40-45 minutes: handover

Collect one explanation from each student. On the planning page, record which step needs reteaching, who requested help, and what was completed. Use neutral observations, not diagnostic labels.

Substitute lesson • Practice

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. A bag holds 2 red, 6 blue, and 4 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

2. A bag holds 7 red, 5 blue, and 3 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

3. A bag holds 3 red, 6 blue, and 4 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

4. A bag holds 6 red, 5 blue, and 7 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.



Substitute lesson • Worked solutions

1. $\frac{1}{6}$; $\frac{5}{6}$

There are 12 equally likely counters. Red: $\frac{2}{12}$; not red: $\frac{10}{12}$. These disjoint events cover all 12 counters.

2. $\frac{7}{15}$; $\frac{8}{15}$

There are 15 equally likely counters. Red: $\frac{7}{15}$; not red: $\frac{8}{15}$. These disjoint events cover all 15 counters.

3. $\frac{3}{13}$; $\frac{10}{13}$

There are 13 equally likely counters. Red: $\frac{3}{13}$; not red: $\frac{10}{13}$. These disjoint events cover all 13 counters.

4. $\frac{1}{3}$; $\frac{2}{3}$

There are 18 equally likely counters. Red: $\frac{6}{18}$; not red: $\frac{12}{18}$. These disjoint events cover all 18 counters.

Emergency substitute lesson

Before students arrive

Locate the class schedule, seating plan, emergency procedures, and approved learner supports from the school. This academic packet does not replace those school-specific instructions.

0-5 minutes: settle

Read the goal aloud: Use equally likely outcomes to model chance. Let students explain a familiar strategy to a partner.

5-15 minutes: model

No copies or devices? Write the first worked example on the board; students use blank paper. List the equally likely outcomes. Probability is favorable outcomes divided by all outcomes. Check that probabilities total one.

15-30 minutes: practice

Distribute the next practice page. Students solve independently first, then compare one explanation in pairs. If copying is unavailable, read one question at a time and allow drawings or oral responses.

30-40 minutes: discuss

Use the following key. Ask students to locate the first step where methods diverged; invite a correction supported by a model.

40-45 minutes: handover

Collect one explanation from each student. On the planning page, record which step needs reteaching, who requested help, and what was completed. Use neutral observations, not diagnostic labels.

Emergency substitute lesson • Practice

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. A bag holds 9 red, 3 blue, and 2 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

2. A bag holds 3 red, 2 blue, and 7 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

3. A bag holds 5 red, 4 blue, and 2 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.

4. A bag holds 5 red, 2 blue, and 3 green counters, identical except for color. One is drawn at random. Find $P(\text{red})$, $P(\text{not red})$, and explain why their sum is 1.



Emergency substitute lesson • Worked solutions

1. $\frac{9}{14}$; $\frac{5}{14}$

There are 14 equally likely counters. Red: $\frac{9}{14}$; not red: $\frac{5}{14}$. These disjoint events cover all 14 counters.

2. $\frac{1}{4}$; $\frac{3}{4}$

There are 12 equally likely counters. Red: $\frac{3}{12}$; not red: $\frac{9}{12}$. These disjoint events cover all 12 counters.

3. $\frac{5}{11}$; $\frac{6}{11}$

There are 11 equally likely counters. Red: $\frac{5}{11}$; not red: $\frac{6}{11}$. These disjoint events cover all 11 counters.

4. $\frac{1}{2}$; $\frac{1}{2}$

There are 10 equally likely counters. Red: $\frac{5}{10}$; not red: $\frac{5}{10}$. These disjoint events cover all 10 counters.

A completed lesson-plan example

Level and focus

Grade 7; NC.7.SP.7; Use equally likely outcomes to model chance.

Evidence of learning

Student represents the quantities, selects a valid method, calculates accurately, and explains the result with appropriate units.

Launch / model

Use worked example 1. Ask students to annotate the role of every quantity. Demonstrate the model before moving to symbols.

Guided / independent practice

Use cards 1-4 in pairs; assign classwork independently. Change grouping according to the explanation you hear rather than speed alone.

Formative check

Use the four-question exit ticket. Score each item 0-2: 2 for an accurate result with coherent reasoning, 1 for a sound method with a minor error or an unexplained correct result, 0 for missing or unsupported reasoning.

Next teaching decision

0-3 points: reteach with a concrete model. 4-6: correct the specific misconception with a short example. 7-8: invite a second representation or a student-created problem. These are local instructional decisions, not mastery cutoffs validated by a test publisher.

Plan your next lesson

Date, class, and focus / exact local standard:

What will students do to show understanding?

Launch and worked example:

Guided practice and independent task:

Access supports and extension:

Exit evidence and next lesson decision:



Access, sources, and classroom license

AIG: Invite two methods, a counterexample, or a new task with a specified solution. Ask students to explain limits of a shortcut.

504: Apply the student's plan: breaks, extra time, enlarged print, reduced copying, or alternate response methods as documented.

EC: Use the IEP to select supports. Model one step, offer counters or a diagram, reduce the visible item set, and accept an oral explanation when appropriate.

ESL: Preview quantity, difference, factor, or other relevant terms. Let students rehearse in a home language, then use "I chose ___ because ___." Maintain the mathematical goal.

Official alignment sources • accessed September 15, 2026

NC.7.SP.7 is verified in the NCDPI mathematics progression. The activity objective is original wording; it describes the skill focus, not the full standard text.

NCDPI mathematics standards:

<https://www.dpi.nc.gov/documents/publications/catalog/ma196-vertical-progression-math/open>

NCDPI EOG mathematics specifications:

<https://www.dpi.nc.gov/documents/accountability/testing/eog/eog-mathematics-grades-3-8-test-specifications>

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